AP Biology Unit 3: 酶相关题目（去重版）

45 题 ｜ 含答案解析 ｜ 重新编号 1–45

编号对照：原 Q95–126 → 新 1–32；原 Q129–141 → 新 33–45

原 Q127–128 为混入的呼吸作用题，已剔除；Q121/Q134/Q137

经二次检查为同材料不同设问，均予保留

1. Acetylcholinesterase (AChE) is a protein that catalyzes the conversion of acetylcholine to acetate and choline. When the concentration of AChE in an aqueous solution is held constant, the rate of the reaction catalyzed by AChE increases with increasing concentrations of substrate. At low concentrations of acetylcholine, a small increase in the substrate concentration results in a large increase in the reaction rate. At high concentrations of acetylcholine, however, a large increase in the substrate concentration results in only a small increase in the reaction rate.

Which of the following statements correctly explains the observed effect of the acetylcholine concentration on the rate of the enzyme-catalyzed reaction?

(A) The active site of AChE is specific for acetylcholine, and only one substrate molecule can occupy the active site at a time. ✓

(B) AChE begins converting product into substrate as the acetylcholine concentration changes from low to high.

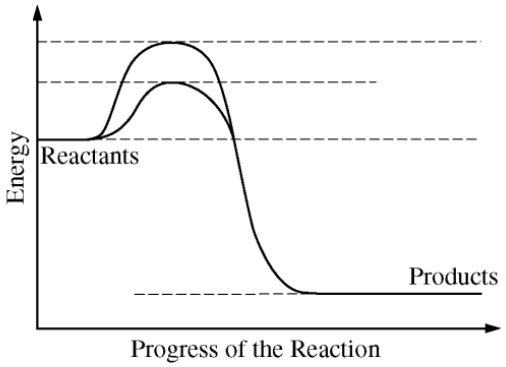
(C) The AChE protein becomes denatured as the acetylcholine concentration changes from low to high.

(D) The substrate specificity of AChE changes as the acetylcholine concentration changes from low to high.

Answer A

Correct. Acetylcholine is the substrate for AChE, and since the shape and charge of the active site is specific for acetylcholine, only one substrate molecule can occupy the active site at a time. Therefore, at high concentrations of acetylcholine, AChE becomes saturated with substrate. When an enzyme is saturated with substrate, adding more substrate to the solution will not lead to an observable increase in the reaction rate.

2. The diagram below shows energy changes in a specific chemical reaction with and without the addition of an enzyme to the reaction.



Which of the following questions can best be answered by the diagram?

(A) Does the addition of an enzyme reduce the activation energy required for a reaction? ✓

(B) Does the addition of an enzyme result in the formation of covalent bonds?

(C) Does the addition of an enzyme produce a greater amount of products?

(D) Does the addition of an enzyme change the pathway for the reaction?

3. A researcher proposes a model of an enzyme-catalyzed reaction in which a reactant is converted to a product. The model is based on the idea that the reactant passes through a transition state within the enzyme-substrate complex before the reactant is converted to the product.

Which of the following statements best helps explain how the enzyme speeds up the reaction?

(A) The enzyme's active site binds to and stabilizes the reactant, which decreases the free-energy change of the reaction.

(B) The enzyme's active site binds to and stabilizes the transition state, which decreases the activation energy of the reaction. ✓

(C) The enzyme's active site binds to and stabilizes the product, which increases the amount of energy released by the reaction.

(D) The enzyme's active site binds to and stabilizes both the reactant and the product at the same time, which increases the reaction's equilibrium constant.

Answer B

Correct. An enzyme typically speeds up a reaction by stabilizing a transition state in the reaction, which decreases the reaction's activation energy. The enzyme's active site binds more tightly to the transition state than to either the substrate or the product, which speeds up the reaction and allows the enzyme to

catalyze the same reaction repeatedly.

4. Amylase is a protein that catalyzes the conversion of starch to simple sugars. Amylase activity in an aqueous solution can be measured by using iodine as a starch indicator. A solution containing iodine and starch will have a dark-blue color, whereas a solution containing iodine but no starch will have a light-brown color. The color change of an iodine solution from dark blue to light brown can be used to measure the rate at which starch is converted to simple sugars.

A student designs an experiment to investigate the effect of environmental pH on amylase function. The design of the experiment is presented in Table 1.

Table 1. An experiment for investigating the effect of pH on amylase function

Test Tube	Environmental pH	Starch Added	Amylase Added	Iodine Added	Pretreated by Boiling
I	5	Yes	Yes	Yes	No
II	6	Yes	Yes	Yes	No
III	7	No	Yes	Yes	No
IV	7	Yes	No	Yes	No
V	7	Yes	Yes	Yes	Yes
VI	7	Yes	Yes	Yes	No
VII	8	Yes	Yes	Yes	No
VIII	9	Yes	Yes	Yes	No

Which of the following statements best justifies the inclusion of test tube V as a control in the experiment?

(A) It will provide a measurement of amylase activity at an acidic pH.

(B) It will provide a measurement of amylase activity at a basic pH.

(C) It will show the color change that occurs in the absence of enzyme activity. ✓

(D) It will show the color change that occurs in the absence of the amylase protein.

Answer C

Correct. Test tube V will contain all the components of the reaction mixture, but the amylase protein will be inactive because it will have been denatured during the pretreatment. Test tube V is an appropriate control in the experiment because it will show the color change that occurs in the absence of amylase activity.

5. A researcher claims that some bacteria contain factors that influence the function of a particular enzyme but other bacteria do not. To test the claim experimentally, the researcher will grow two different bacterial strains in separate liquid cultures and isolate the contents of the cells in each culture. The researcher will add different combinations of cellular contents, substrate, and enzyme to test tubes containing a buffer solution adjusted to the optimal pH of the enzyme and then measure the rate of product formation. The design of the researcher's experiment is presented in Table 1.

Table 1. An experiment for investigating the influence of bacterial factors on enzyme function

Test Tube	Source of Cellular Contents	Cellular Contents Added	Substrate Added	Enzyme Added
1	Strain 1	No	Yes	Yes
2	Strain 1	Yes	No	Yes
3	Strain 1	Yes	Yes	No
4	Strain 1	Yes	Yes	Yes
5	Strain 2	No	Yes	Yes
6	Strain 2	Yes	No	Yes
7	Strain 2	Yes	Yes	No
8	Strain 2	Yes	Yes	Yes

Which of the following statements best justifies the inclusion of test tubes 3 and 7 in the experiment?

(A) They will show whether the isolated cellular contents have enzymatic activity. ✓

(B) They will show whether environmental pH affects the function of the enzyme.

(C) They will show the rate of product formation in the absence of bacterial factors.

(D) They will show the rate of product formation in the absence of the substrate.

Answer A

Correct. The researcher will add cellular contents and the substrate but not the enzyme to test tubes 3 and 7. As such, test tubes 3 and 7 will show whether the cellular contents have enzymatic activity.

6. A researcher designs an experiment to investigate the effect of environmental temperature on the function of an enzyme. For each trial included in the experiment, the researcher will add the enzyme and its substrate to an aqueous buffer solution and then measure the amount of product formed over 20 minutes.

Which of the following must remain the same for all trials of this experiment?

(A) The initial concentration of the substrate ✓

(B) The final concentration of the product

(C) The three-dimensional structure of the enzyme

(D) The temperature of the aqueous buffer solution

Answer A

Correct. Keeping the initial substrate concentration constant allows the researcher to compare the reaction rates under different environmental conditions. The reaction rates will provide the researcher with information about the enzyme's function in different environments.

7. Catalase is an enzyme found in yeast cells that facilitates the chemical breakdown of hydrogen peroxide to water and oxygen gas. An experiment was conducted to determine the effect of pH on catalase function. Five buffer solutions of varying pH (2, 4, 6, 8, and 10) were prepared and added to separate test tubes. Hydrogen peroxide (H_2O_2) was added to each test tube. Yeast was added, and the reactions were timed. After 1 minute the amount of oxygen gas released was determined by measuring the foam layer produced in each test tube. Figure 1 illustrates the experimental setup.

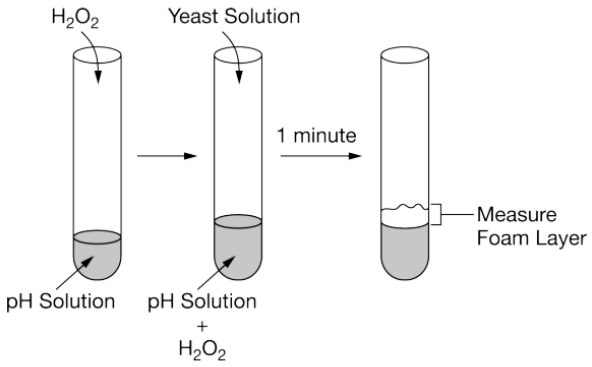


Figure 1. Illustration of experimental procedure

A set of five additional test tubes were prepared and used as controls. Which of the following best describes the contents expected to be contained in one of the five control test tubes?

(A) pH 4 buffer solution and hydrogen peroxide only ✓

(B) All five pH buffer solutions combined and hydrogen peroxide only

(C) Water, hydrogen peroxide, and yeast

(D) Water and yeast only

Answer A

Correct. Since the experiment is studying the effect of pH on catalase in yeast, the control set should determine what would happen without yeast present. Each test tube in the control set should contain hydrogen peroxide and one of the pH solutions.

For following group of questions first study the description of the data and then choose the one best answer to each question following it and fill in the corresponding oval on the answer sheet.

To study the actions of the enzyme catalase on hydrogen peroxide, students performed the following experiment. Catalase was extracted from potatoes by blending raw potatoes in a blender with cold distilled water. The filtrate was stored on ice. The following hydrogen peroxide solutions were made: 1 percent, 5 percent, 10 percent, and 15 percent. Filter paper disks were soaked in the catalase filtrate and dropped into beakers containing the various solutions. The activity of the enzyme was measured by the amount of time it took for the disks to float to the surface of the solution on the bubbles produced by the reaction. The following data were obtained.

Hydrogen Peroxide Solution	Average Time, in Seconds, for Disks to Float
1%	30 sec
5%	25 sec
10%	20 sec
15%	10 sec

8. Which of the following experimental designs should the students use as a control for the experiment?

(A) Place a catalase-soaked disk in a beaker of water. ✓

(B) Drop the disks from different heights into the solution.

(C) Poke the floating disks back down to the bottom of the beaker and retime the rise of the disks.

(D) Shake the beakers during the time the disks are rising.

(E) Dip the disks in the 5% solution before putting them in any other solution.

9. Which of the following best describes why ice was used during this experiment?

(A) To increase the activity of the enzyme

(B) To retard the breakdown of the catalase ✓

(C) To purge the solution of excess air trapped during blending

(D) To slow the catalase molecules to increase the probability of contact with the hydrogen peroxide molecules

(E) To increase the size of the active site on the enzyme

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10. Which of the following best explains why the disks rose to the surface faster in the more concentrated hydrogen peroxide solutions?

(A) There was more enzyme present in the more concentrated solutions.

(B) A greater amount of heat was generated in the more concentrated solutions.

(C) The more concentrated solutions lowered the activation energy of the reaction.

(D) The higher substrate concentrations in the more concentrated solutions speeded the reaction. ✓

(E) The density of the water was lower in the more concentrated solutions.

11. If the potato solution was boiled for 10 minutes and cooled for 10 minutes before being tested, the average time for the disks to float to the surface of the hydrogen peroxide solution would be

(A) less than 1 second

(B) 5 seconds

(C) 10 seconds

(D) 30 seconds

(E) more than 30 seconds ✓

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12. Which of the following statements best helps explain the reaction specificity of an enzyme?

(A) The free energy of the reactants is greater than the free energy of the products.

(B) The equilibrium constant of the reaction is much greater than 1.

(C) The shape and charge of the substrates are compatible with the active site of the enzyme. ✓

(D) The concentration of the enzyme inside living cells is greater than the concentration of substrate.

Answer C

Correct. The reaction specificity of an enzyme is determined by the compatibility of the substrate with the enzyme's active site. A substrate's shape and charge affect how it fits into an enzyme's active site and whether it forms favorable interactions with the enzyme.

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13. Gelatin is a protein that is derived from collagen which is found in the bones, skin, and connective tissue of animals. To investigate the ability of various enzymes to digest gelatin, a group of students set up an assay involving camera film. Camera film contains gelatin and appears black when exposed to light but turns clear as the gelatin gets broken down. The students incubated pieces of exposed camera film in test tubes, each containing one of three different enzyme solutions (trypsin, lipase, or amylase) as indicated in Figure 1. The students recorded the time it took for the enzymes to digest the gelatin in each test tube, turning the film from black to clear.

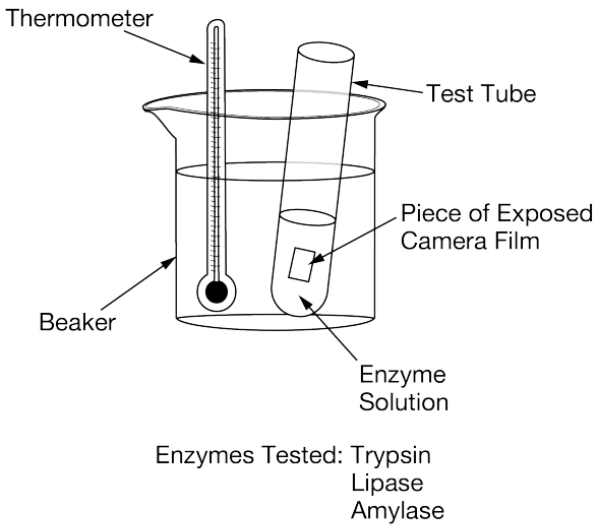


Figure 1. Diagram of experimental setup.

Which of the following would be the most appropriate control for this experiment?

(A) A test tube containing no camera film

(B) A test tube containing only a piece of exposed camera film

(C) A test tube containing a piece of exposed camera film submerged in water ✓

(D) A test tube containing a piece of exposed camera film and all three enzyme solutions

Answer C

Correct. Having one test tube with only water and camera film will show that water is not the substance that causes the film to change color.

14. The enzyme hexokinase catalyzes the conversion of glucose to glucose-6-phosphate, which is an important step in glycolysis. The reaction involves the transfer of a phosphate group from ATP to glucose.

Either a glucose molecule or a water molecule can fit in the active site of hexokinase. The presence of a water molecule in hexokinase's active site would result in the hydrolysis of ATP to ADP instead of the conversion of glucose to glucose-6-phosphate.

Which of the following statements best helps explain the reaction specificity of hexokinase?

(A) Both glucose and water are polar molecules that form favorable interactions with charged and polar amino acid side chains in hexokinase's active site.

(B) Both glucose and water have oxygen atoms that can form covalent bonds with the phosphorus atoms of phosphate groups.

(C) Glucose is an energy-rich organic molecule that can be broken down by glycolysis to produce ATP, whereas water is an inorganic molecule.

(D) Glucose has the right shape and charge to cause hexokinase to undergo a structural change needed for catalysis, whereas water does not. ✓

Answer D

Correct. Hexokinase catalyzes the conversion of glucose to glucose-6-phosphate but not the hydrolysis of ATP. Hexokinase's reaction specificity is explained by the compatibility of glucose and hexokinase's active site, which results in part from glucose's shape and charge. Because of this compatibility, hexokinase undergoes a structural change when it binds to glucose but not to water, and the structural change is necessary for hexokinase to become catalytically active.

15. A student designs an experiment to investigate the influence of temperature on enzyme function. The student's plan is presented in Table 1.

Table 1. An experiment to investigate the influence of temperature on enzyme function

Test Tube	Substrate Added	Enzyme Added	Temperature (°C)
1	Yes	No	10
2	Yes	Yes	10
3	Yes	No	20
4	Yes	Yes	20
5	Yes	No	30
6	Yes	Yes	30
7	Yes	No	40
8	Yes	Yes	40

Which test tubes are controls in the experiment?

(A) Test tubes 1 and 2 only

(B) Test tubes 5 and 6 only

(C) Test tubes 1, 3, 5, and 7 ✓

(D) Test tubes 2, 4, 6, and 8

Answer C

Correct. The test tubes that do not contain the enzyme are negative controls in the experiment. The negative controls help confirm that the results of the experiment are related to enzyme function and not some other factor.

16. Aminolevulinate dehydratase (ALAD) is an enzyme that relies on zinc as a cofactor. A zinc ion binds to the ALAD active site, where it forms favorable interactions with the side chains of three amino acids. Researchers have found that substituting a lead ion for a zinc ion in the ALAD active site causes inhibition of ALAD.

Which of the following statements best helps explain how the lead ion causes inhibition of ALAD?

(A) It changes the shape and charge of the substrate so that it becomes more compatible with ALAD's active site.

(B) It changes the amino acid sequence of the ALAD protein so that the enzyme catalyzes a different reaction.

(C) It changes the three-dimensional structure of the active site so that ALAD is no longer compatible with its substrate. ✓

(D) It changes the enzyme-substrate complex so that the transition state is more stable and the reaction proceeds at a faster rate.

Answer C

Correct. The substitution of a lead ion for a zinc ion most likely changes the three-dimensional structure of the ALAD active site. The three-dimensional structure of the active site plays an important role in the compatibility between an enzyme and its substrate. Changing the structure of the active site will likely interfere with enzyme function by disrupting the enzyme-substrate interaction.

17. Pectinase is a protein that catalyzes the breakdown of pectic polysaccharides in plant cell walls. A researcher designs an experiment to investigate the effect of salinity on the ability of pectinase to lower the activation energy of the reaction involved. The design of the experiment is presented in Table 1. For each test tube, the researcher will measure the amount of product formed over 20 minutes.

Table 1. An experiment to investigate the effect of salinity on pectinase function

Test Tube	Sodium Chloride Concentration (molar)	Temperature (degrees C)	Substrate Added	Pectinase Added
1	0	23	Yes	No
2	0	23	Yes	Yes
3	0.5	23	Yes	No
4	0.5	23	Yes	Yes
5	1.0	23	Yes	No
6	1.0	23	Yes	Yes
7	1.5	23	Yes	No
8	1.5	23	Yes	Yes

Which of the following statements best helps justify the inclusion of test tube 5 in the experiment?

(A) It will act as a control for test tube 4 by showing the effect of the presence or absence of the substrate.

(B) It will act as a control for test tube 4 by showing the effect of a change in environmental temperature.

(C) It will act as a control for test tube 6 by showing the effect of the presence or absence of the enzyme. ✓

(D) It will act as a control for test tube 6 by showing the effect of a change in sodium chloride concentration.

Answer C

Correct. Test tube 5 will act as a control for test tube 6. The solutions in test tubes 5 and 6 are nearly identical except that the researcher will add pectinase to test tube 6 but not to test tube 5. A comparison of the data for test tubes 5 and 6 will show the effect of the presence or absence of the enzyme.

Researchers investigated the influence of environmental pH on the activity of peroxidase, an enzyme that catalyzes the conversion of hydrogen peroxide to water and oxygen gas. In an experiment, the researchers added a hydrogen peroxide solution containing guaiacol to several identical test tubes and adjusted the solution in each test tube to a different pH. The researchers included the guaiacol because it caused the solutions to change color as the reactions proceeded, which the researchers relied on for measuring reaction rates. Finally, the researchers added the same amount of peroxidase to each test tube and measured the rate of each reaction at $23\,^{\circ}\mathrm{C}$. The results of the experiment are represented in Figure 1.

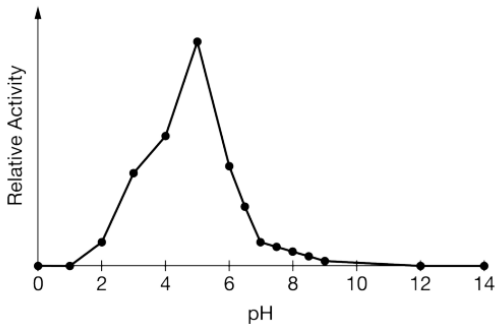


Figure 1. The effect of pH on peroxidase activity

18. One of the researchers proposes using oxygen gas production to measure reaction rates. Which of the following statements best justifies the use of the proposed modification as a way of creating an appropriate control for the investigation?

(A) The experiment can be repeated without hydrogen peroxide, which will help eliminate an uncontrolled variable.

(B) The experiment can be repeated without peroxidase, which will introduce a second independent variable.

(C) The experiment can be repeated without guaiacol, which will reveal the effect of guaiacol on the reaction rates. ✓

(D) The experiment can be repeated without water, which will reveal whether the reaction can occur inside a living cell.

Answer C

Correct. In the original experiment, the researchers used guaiacol as a reaction indicator. By using oxygen gas production to measure reaction rates, the researchers no longer need the guaiacol and can repeat the experiment without it, which will reveal the effect of guaiacol on the reaction rates.

19. Which of the following actions will provide a negative control for the investigation?

(A) Repeating the experiment at $25^{\circ}C$

(B) Repeating the experiment using twice the amount of hydrogen peroxide

(C) Repeating the experiment using twice the amount of peroxidase

(D) Repeating the experiment using heat-denatured peroxidase ✓

Answer D

Correct. A negative control is a group in which no response is expected. It is the opposite of the positive control, in which a known response is expected. Heat-denatured peroxidase will be inactive. Repeating the experiment using heat-denatured peroxidase will allow the researcher to measure reaction rates in the presence of inactive peroxidase.

20. Based on Figure 1, which of the following statements best predicts the effect that a change from a moderately acidic environment (pH near 6) to a basic environment will have on peroxidase activity?

(A) Peroxidase activity will decrease. ✓

(B) Peroxidase activity will increase.

(C) Peroxidase activity will stay the same.

(D) Peroxidase activity will increase at first and then decrease.

Answer A

Correct. Based on Figure 1, peroxidase's optimal pH is 5. A change from a moderately acidic environment (pH near 6) to a basic environment (pH above 7) will result in a decrease in peroxidase activity.

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21. A researcher designs an experiment to investigate whether soil bacteria trigger the synthesis of defense enzymes in plant roots. The design of the experiment is presented in Table 1. For each group in the experiment, the researcher will determine the average rate of change in the amount of defense enzymes in the roots of the seedlings.

Table 1. An experiment to investigate the effect of soil bacteria on plant defenses

Group	Number of Seedlings	Type of Soil	Treatment Solution
1	10	Sterile potting soil	Contains actively reproducing soil bacteria
2	10	Sterile potting soil	Contains heat-killed soil bacteria
3	10	Sterile potting soil	Contains no soil bacteria

Which of the following statements best helps justify the inclusion of group 2 as one of the controls in the experiment?

(A) It will show whether the changes observed in group 1 depend on the metabolic activity of soil bacteria. ✓

(B) It will show whether the changes observed in group 1 depend on the type of plants used in the experiment.

(C) It will show the average growth rate of seedlings that are maintained in a nonsterile environment.

(D) It will show the changes that occur in the roots of seedlings following an infection by soil bacteria.

Answer A

Correct. The seedlings in group 1 will be treated with a solution that contains actively reproducing bacteria, which will be metabolically active. In contrast, the seedlings in group 2 will be treated with a solution that contains heat-killed bacteria, which will be metabolically inactive. The researcher will be able to compare the results for group 1 with those for group 2. The comparison will help the researcher determine whether the changes observed in group 1 depend on the metabolic activity of soil bacteria.

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22. A researcher claims that increasing the environmental temperature by $10^{\circ}C$ will double the rate of an enzyme-catalyzed reaction. To test the claim, the researcher designs an experiment that uses a particular enzyme isolated from plants. The design of the experiment is presented in Table 1. For each test tube in the experiment, the researcher will measure the rate of product formation.

Test Tube	Temperature ($^{\circ}C$)	Substrate Added	Enzyme Added	Pretreated by Boiling
1	15	Yes	Yes	No
2	25	No	Yes	No
3	25	Yes	No	No
4	25	Yes	Yes	No
5	25	Yes	Yes	Yes
6	35	Yes	Yes	No

Which of the following statements best helps justify the inclusion of test tube 5 as a control in the experiment?

(A) It will provide a measurement of product formation in the absence of the substrate.

(B) It will provide a measurement of product formation in the presence of a denatured enzyme. ✓

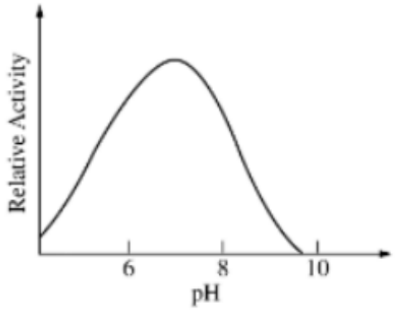
(C) It will show the effect of doubling the amount of substrate on the rate of product formation.

(D) It will show the effect of increased enzyme activity on the rate of product formation.

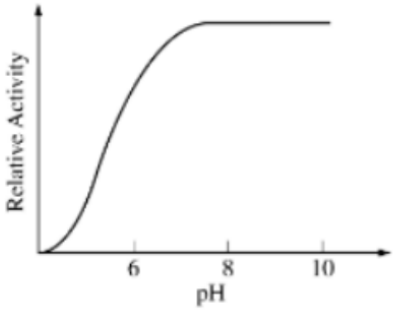
Answer B

Correct. The contents of test tube 5 will be pretreated by boiling. The pretreatment will denature the plant enzyme used in the experiment. As such, test tube 5 will provide a measurement of product formation in the presence of a denatured enzyme.

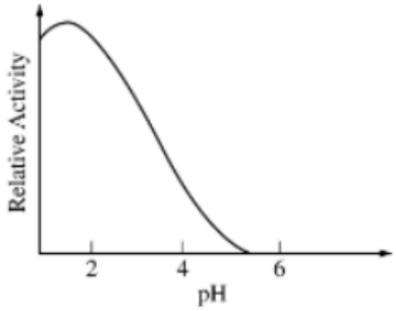
For the following questions: Graphs I-IV depict the effect of pH on the activity of four different hydrolytic enzymes.



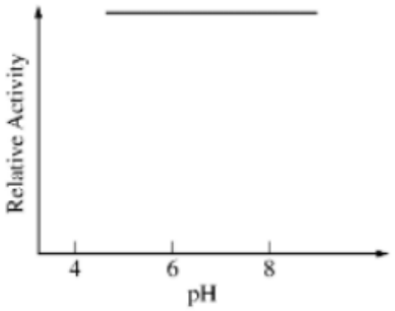
Graph I



Graph II



Graph III



Graph IV

23. Graphs representing enzymes sensitive to changes in pH include which of the following?

(A) I only
(B) IV only
(C) II and III only
(D) III and IV only
(E) I, II, and III only ✓

24. The most likely explanation for the results shown in Graph I is that

(A) pH affects the shape of the active site of the enzyme ✓
(B) pH affects the temperature of the reaction
(C) the enzyme has a quaternary structure
(D) the enzyme has disulfide bonds
(E) pH affects the primary structure of the enzyme

25. Enzymes with their highest activity at an alkaline (basic) pH are represented by which of the following graphs?

(A) I only

(B) II only ✓

(C) III only

(D) I and III only

(E) I and IV only

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26. In an experiment, a researcher prepares a reaction mixture by dissolving a substance in a buffered solution. The substance is the substrate of a certain enzyme. The researcher adds a small amount of the enzyme to the reaction mixture and measures the amount of product that is formed over time. The data are represented in Figure 1.

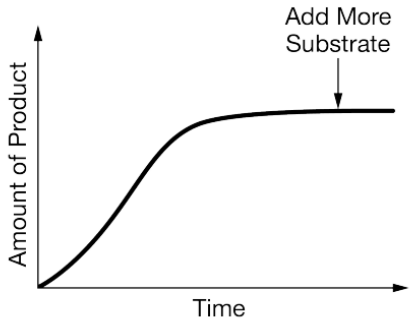


Figure 1. The amount of product formed by an enzyme-catalyzed reaction over time

Which of the following best predicts the immediate result of adding more substrate to the reaction mixture at the point indicated by the arrow in Figure 1?

(A) The amount of product will decrease until the reaction rate goes to zero.

(B) The amount of product will decrease until the reaction reaches its equilibrium point or until the enzyme is been used up by the reaction.

(C) The amount of product will increase until the reaction reaches its equilibrium point or until the substrate is used up by the reaction. ✓

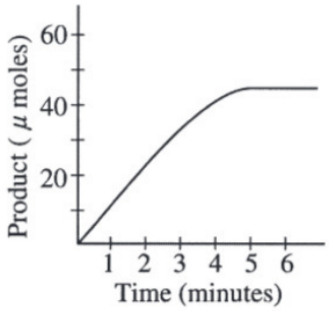
(D) The amount of product will increase without stopping because the enzyme will be unchanged by the reaction.

Answer C

Correct. The graph shown in Figure 1 indicates that the amount of product became constant as a consequence of the reaction reaching its equilibrium point or as a consequence of the substrate being used up. Adding more substrate will likely result in an increase in the amount of product because the reaction catalyzed by the enzyme will convert the added substrate into product.

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27. A scientist determined the rate of an enzyme-catalyzed reaction by measuring the amount of product formed over time. The following curve was generated from the data collected.



During which time interval is the reaction rate lowest?

(A) 0-1 minute

(B) 1-2 minutes

(C) 2-3 minutes

(D) 3-4 minutes

(E) 4-5 minutes ✓

The coral species *Acropora millepora* is a marine animal that has a symbiotic relationship with algae species in the genus *Symbiodinium*. An increase in the concentration of dissolved CO_2 in oceans has a negative impact on this relationship and the health of the coral species. Scientists investigated the effect of CO_2 concentrations on the number of algae cells found in the coral (Figure 1) and on both the gross (total) and net rates of photosynthesis of those algae (Figure 2). The net rate of photosynthesis is equal to the gross rate of photosynthesis minus the rate of respiration by the algae.

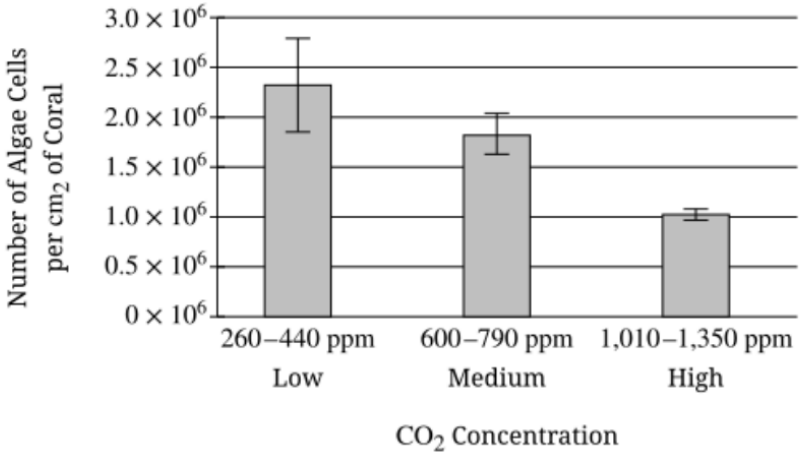


Figure 1. Effect of CO_2 concentration on the number of algae per cm^2 of coral. Error bars represent $\pm 2SE_{\bar{x}}$.

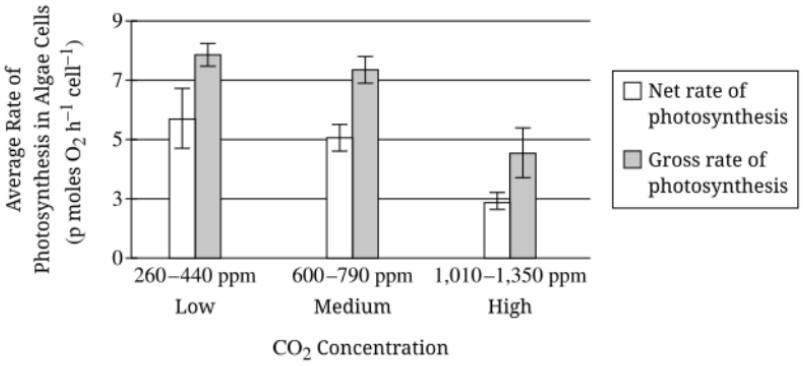


Figure 2. Average gross and net rates of photosynthesis in algae under various concentrations of CO_2. Error bars represent $\pm 2SE_{\bar{x}}$.

28. Researchers observe that the water with the high CO_2 concentration has a lower pH than does the water with the low CO_2 concentration. The researchers claim that lower pH alters the structure of the enzymes of photosynthesis.

Which of the following most likely explains how a lower pH could alter the structure of photosynthetic enzymes?

(A) There will not be enough H^+ ions for the ribosome to correctly form peptide bonds between amino acids during synthesis of the enzymes.

(B) There will not be enough H^+ ions for the necessary hydrogen bonds to form in the secondary, tertiary, and quaternary structures of the enzymes.

(C) There will be additional H^+ ions that can disrupt peptide bonds in the primary structure of the enzymes.

(D) There will be additional H^+ ions that can disrupt hydrogen bonds in the secondary, tertiary and quaternary structures of the enzymes. ✓

Answer D

Correct. Additional H^+ ions compete for hydrogen bonding with amino acid side chains and therefore denature the protein and alter the structure of the photosynthetic enzymes.

29. Based on Figure 2, which of the following best supports the claim that CO_2 concentration and photosynthetic productivity are inversely related?

(A) The net rate of photosynthesis is less than the gross rate of photosynthesis at the high concentration of CO_2.

(B) Both the net and gross rates of photosynthesis at the high concentration of CO_2 are lower than those at the low concentration of CO_2. ✓

(C) The net rate of photosynthesis at the low concentration of CO_2 is lower than the gross rate of photosynthesis at the high concentration of CO_2.

(D) The difference between net and gross rates of photosynthesis is greater at the low concentration of CO_2 than it is at the high concentration of CO_2.

Answer B

Correct. This describes an inverse relationship where one measure increases as the other decreases. Because the rate of photosynthesis decreases as CO_2 concentration increases, this supports the claim that CO_2 concentration is inversely related to photosynthetic rate (productivity).

30. Alcohol dehydrogenase (ADH) is an enzyme that aids in the decomposition of ethyl alcohol ($\mathrm{C_2H_5OH}$) into nontoxic substances. Methyl alcohol acts as a competitive inhibitor of ethyl alcohol by competing for the same active site on ADH. When attached to ADH, methyl alcohol is converted to formaldehyde, which is toxic in the body.

Which of the following statements best predicts the effect of increasing the concentration of substrate (ethyl alcohol), while keeping the concentration of the inhibitor (methyl alcohol) constant?

(A) There will be an increase in formaldehyde because ADH activity increases.

(B) Competitive inhibition will be terminated because ethyl alcohol will bind to methyl alcohol and decrease ADH activity.

(C) The peptide bonds in the active site of the enzyme will be denatured, inhibiting the enzyme.

(D) Competitive inhibition will decrease because the proportion of the active sites occupied by substrate will increase. ✓

Answer D

Correct. The increase in amount of substrate will increase the number of enzyme-ethanol complexes formed, increasing the amount of nontoxic product formed from ethyl alcohol.

31. A researcher proposes a model to explain how enzyme-substrate interactions determine enzyme specificity. The model is based on the idea that substrate molecules form favorable interactions with the amino acid side chains in an enzyme's active site.

Based on the model, which of the following statements best explains an enzyme's specificity for a particular substrate molecule?

(A) A hydrophilic molecule interacts with nonpolar side chains in the enzyme's active site.

(B) A hydrophobic molecule interacts with polar side chains in the enzyme's active site.

(C) A molecule with positive charges interacts with positively charged side chains in the enzyme's active site.

(D) A molecule with negative charges interacts with positively charged side chains in the enzyme's active site. ✓

Answer D

Correct. Because opposite charges attract, a molecule with negative charges will form favorable interactions with positively charged side chains in an enzyme's active site. Based on the model, the favorable interactions helps explain the enzyme's specificity for the molecule as a substrate.

32. A researcher claims that different enzymes exhibit maximal function over different pH ranges. To test the claim, the researcher carries out an experiment that includes three different enzymes: pepsin, salivary amylase, and arginase. The results of the experiment are represented in Figure 1.

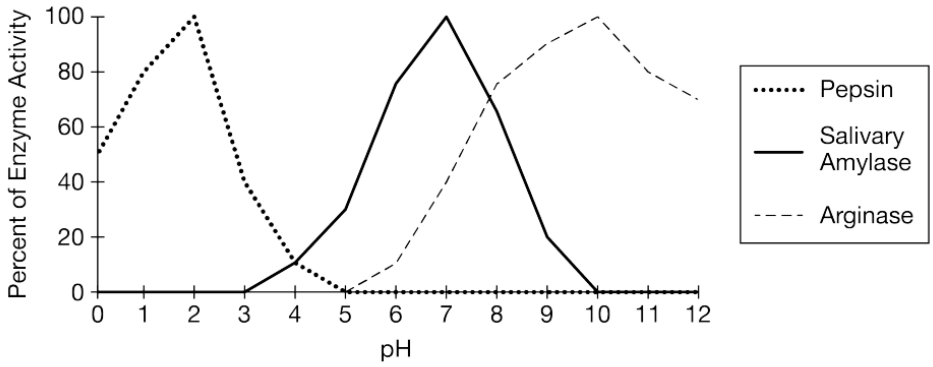


Figure 1. The effect of pH on three different enzymes

Which of the following actions will provide the most appropriate negative control for the experiment?

(A) Repeating the experiment with a fourth enzyme

(B) Repeating the experiment at several different temperatures

(C) Repeating the experiment with denatured enzymes ✓

(D) Repeating the experiment using several different methods for measuring pH

Answer C

Correct. Repeating the experiment with denatured enzymes will provide an appropriate negative control for the experiment. The negative control will show that the changes observed in the experiment result from changes in enzyme function.

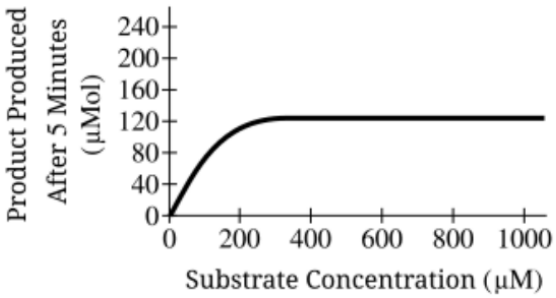
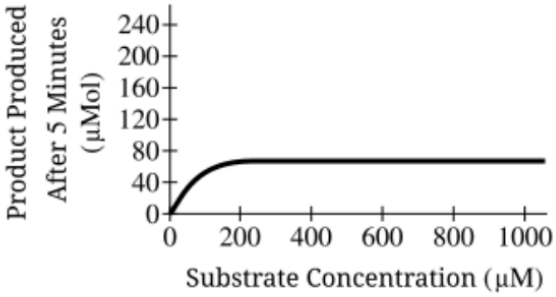


Figure 1. Rate of an enzyme-catalyzed reaction

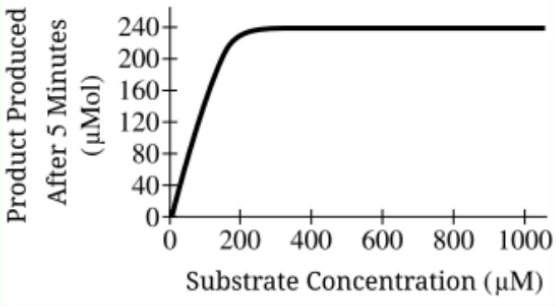
Researchers investigated the dynamics of a reaction catalyzed by an enzyme. The researchers prepared a series of reactions, each with the same concentration of enzyme but with different concentrations of substrate. The researchers measured the amount of product in each reaction mixture after 5 minutes. The results are shown in Figure 1.

33. Which of the following graphs predicts the most likely rate of the enzyme-catalyzed reaction if the concentration of the enzyme in the mixture were doubled?

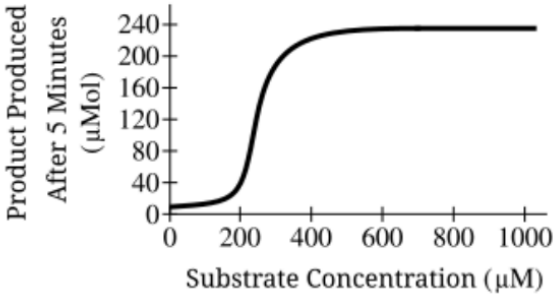
(A)


Product Produced After 5 Minutes (µMol)
240
200
160
120
80
40
0
0
200
400
600
800
1000
Substrate Concentration (µM)

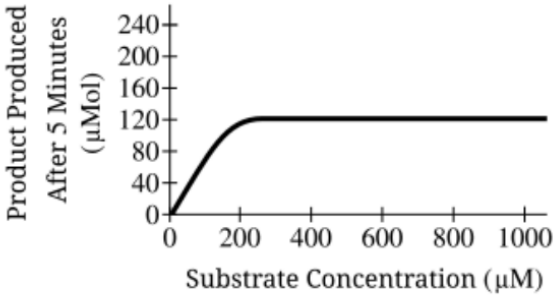
(B)


Product Produced After 5 Minutes (µMol)
240
200
160
120
80
40
0
0
200
400
600
800
1000
Substrate Concentration (µM)

(C)


Product Produced After 5 Minutes (µMol)
240
200
160
120
80
40
0
0
200
400
600
800
1000
Substrate Concentration (µM)

(D)


Product Produced After 5 Minutes (µMol)
240
200
160
120
80
40
0
0
200
400
600
800
1000
Substrate Concentration (µM)

Answer B

Correct. The rate of the reaction is initially proportional to the concentration of enzyme present. When all active sites are occupied, the rate remains constant at a maximum. Thus, doubling the enzyme concentration should double the rate of the reaction at every substrate concentration, and the maximum rate should also double.

34. Which of the following questions best addresses whether a particular inhibitor is competitive or noncompetitive?

(A) Does the inhibitor bind more tightly to the substrate or to the enzyme?

(B) Can the inhibitor be converted to product by the enzyme?

(C) How much inhibitor is necessary to completely inhibit the activity of the enzyme?

(D) Does the inhibitor bind to an allosteric site or the active site of the enzyme? ✓

Answer D

Correct. Competitive enzyme inhibitors affect the rate of reaction by competing with the substrate for binding to the active site of the enzyme. Noncompetitive inhibitors affect the rate of reaction by binding to an allosteric site of the enzyme, altering the shape and thus the function of the enzyme.

35. Which of the following experimental designs would best allow researchers to investigate the effect of pH on the rate of the enzyme-catalyzed reaction?

(A) Measure reaction rates at different pH levels when the enzyme concentration remains the same and the substrate concentration is 100 µM. ✓

(B) Measure the pH of the reaction mixture when the enzyme concentration remains the same and substrate concentrations vary between 0 µM and 1,000 µM.

(C) Measure reaction rates when the enzyme concentration remains the same but the pH of the reaction mixture increases and the substrate concentration increases from 0 µM to 1,000 µM.

(D) Measure reaction rates when the substrate concentration is 100 µM and both the pH of the reaction mixture and the enzyme concentration are increased.

Answer A

Correct. To investigate the role of pH on the reaction rate, the change in pH must be the only independent variable.

36. Which of the following best describes how an increase in the concentration of substrate in the reaction mixture affected the frequency of enzyme-substrate interactions?

(A) The frequency of enzyme-substrate interactions decreased over all substrate concentrations.

(B) The frequency of enzyme-substrate interactions increased exponentially over all substrate concentrations.

(C) The frequency of enzyme-substrate interactions increased until all of the active sites were interacting with substrate. ✓

(D) The frequency of enzyme-substrate interactions increased until all of the enzyme was depleted.

Answer C

Correct. The frequency of enzyme-substrate interactions is determined by the maximum number of enzyme active sites available to bind substrate and convert it to product. The rate of the reaction increases until the active sites of all enzyme molecules are filled, and then the rate of the reaction remains constant.

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37. Which of the following can be used to determine the rate of enzyme-catalyzed reactions?

(A) Rate of disappearance of the enzyme

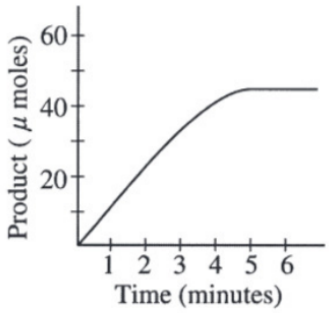
(B) Rate of disappearance of the substrate ✓

(C) Rate of disappearance of the product

(D) Change in volume of the solution

(E) Increase in activation energy

38. A scientist determined the rate of an enzyme-catalyzed reaction by measuring the amount of product formed over time. The following curve was generated from the data collected.



What is the most likely explanation for the change in the slope of the line between 3 and 5 minutes?

(A) The enzyme had denatured.

(B) The enzyme had achieved its maximum velocity.

(C) A large amount of the substrate had been consumed. ✓

(D) An allosteric inhibitor appeared.

(E) There was a dramatic change in the pH.

39. The enzyme trypsin aids in protein digestion in the small intestine. The relative activity of trypsin at different pH values is shown in Figure 1.

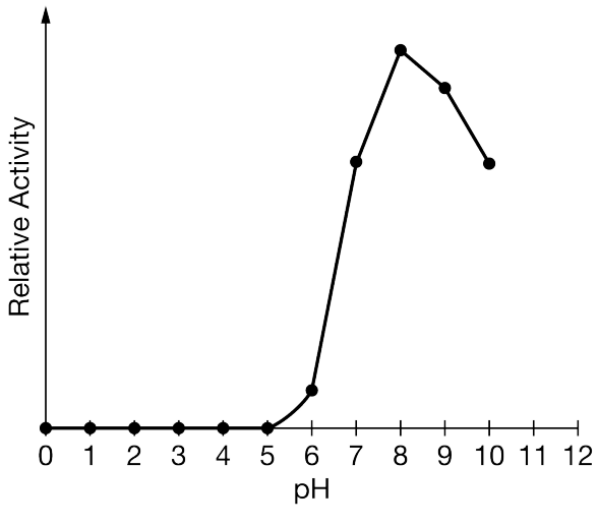


Figure 1. Effect of pH on the activity of trypsin

Which of the following statements best explains the activity levels of trypsin shown in Figure 1 ?

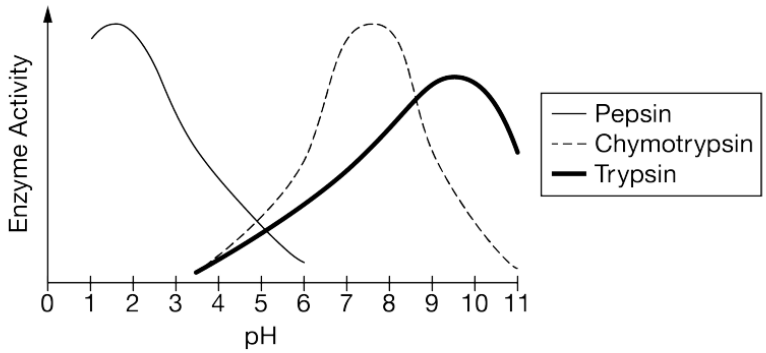
(A) The small intestine releases inhibitor molecules that block the activity of trypsin unless it is at its optimum pH.

(B) The number of effective collisions between trypsin and its substrate increase at higher pH values.

(C) As pH values increase, the substrate concentration decreases, leading to an eventual decline in the rate of the trypsin-catalyzed reaction.

(D) At extremely low pH values, trypsin is denatured and cannot function efficiently. ✓

40. Protein digestion in humans is primarily carried out by three enzymes. Pepsin is found in the stomach $(\mathrm{pH}\ 2)$, where it aids in the breakdown of large proteins into smaller peptides, while trypsin and chymotrypsin are found in the small intestine $(\mathrm{pH}\ 8)$, where they aid in the further breakdown of the proteins into amino acids and dipeptides that can be absorbed into the bloodstream. Graph 1 shows the effect of pH on the activity levels of the three enzymes.



Graph 1. Relative activity of pepsin, trypsin, and chymotrypsin at pH 0 through 11

Which of the following best predicts how the structure and function of pepsin will change as it enters the small intestine?

(A) Pepsin will not change in shape and will continue to break down proteins in the small intestine.

(B) Pepsin will not change in shape but may not work due to the basic environment of the small intestine.

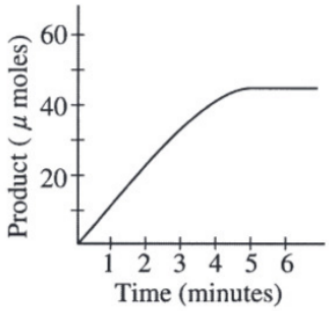
(C) Pepsin will change in shape because of the basic environment of the small intestine; therefore, its enzymatic activity will decrease. ✓

(D) Pepsin will change in shape because of the presence of trypsin and chymotrypsin in the small intestine, both of which act as competitive inhibitors.

Answer C

Correct. The significant change in pH will disturb the hydrogen bonding and cause denaturation of pepsin. As a result, pepsin will stop working in the small intestine.

41. A scientist determined the rate of an enzyme-catalyzed reaction by measuring the amount of product formed over time. The following curve was generated from the data collected.



The rate of the reaction could also be determined by

(A) measuring the change in the amount of enzyme

(B) measuring the change in the amount of substrate ✓

(C) measuring the change in salt concentration

(D) adding more substrate

(E) adding more enzyme

42. Researchers investigated the effect of urea on the three-dimensional structure of a certain enzyme. The researchers dissolved the enzyme in an aqueous buffer solution and added urea to the solution. The enzyme did not appear to have a secondary or tertiary structure. The researchers carefully removed the urea from the solution and determined that the enzyme had the original secondary and tertiary structure again.

Based on the results of the experiment, which of the following statements best predicts the effect of urea on the enzyme's function?

(A) Function will be disrupted by adding the urea and regained by removing the urea. ✓

(B) Function will be disrupted by adding the urea, but it will not be regained by removing the urea.

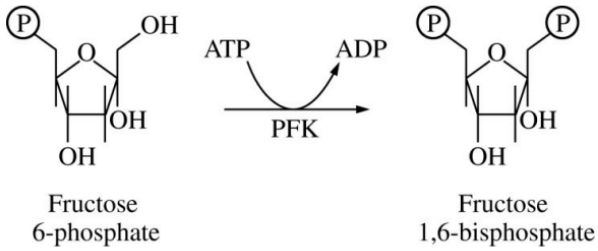
(C) Function will be gained by adding the urea and disrupted by removing the urea.

(D) Function will be unaffected by the addition and removal of the urea.

Answer A

Correct. Based on the information, adding the urea denatures the enzyme, which will disrupt the enzyme's function by disrupting the active site. In contrast, removing the urea causes the enzyme to refold, which will restore the active site and the enzyme's function.

43.



Fructose 6-phosphate

Fructose 1,6-bisphosphate

Figure 1. Reaction catalyzed by phosphofructokinase (PFK) during glycolysis

Phosphofructokinase (PFK) is an enzyme that catalyzes the conversion of fructose 6-phosphate to fructose 1,6-bisphosphate during glycolysis, as represented in Figure 1.

PFK can be allosterically inhibited by ATP at high concentrations. Which of the following is the benefit of regulating glycolysis by the concentration of ATP?

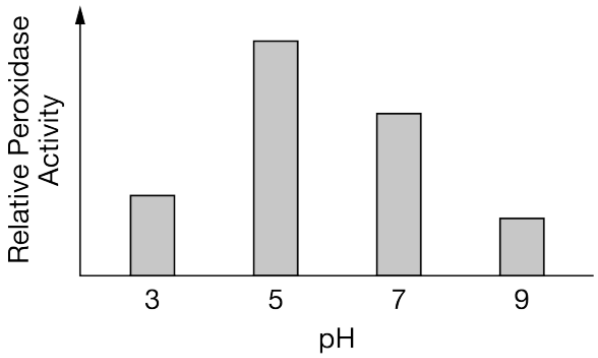
(A) Glycolysis proceeds when the intracellular concentration of ATP is low, which provides ATP to drive cellular reactions. ✓

(B) Glycolysis proceeds when the intracellular concentration of ATP is high and the cell stores ATP for future use.

(C) Glycolysis is inhibited when the intracellular concentration of ATP is low because PFK requires ATP as a substrate for the reaction it catalyzes.

(D) Glycolysis is inhibited when the intracellular concentration of ATP is high because ATP will compete with fructose 1,6-bisphosphate for binding to the active site on the enzyme.

44. The enzyme peroxidase is found in many organisms. It catalyzes the breakdown of hydrogen peroxide into water and oxygen gas. The rate of peroxidase activity at different pH values was assessed by students in the lab. The students' results are shown in graph 1.



Graph 1. Effect of pH on peroxidase activity

If the experiment is repeated at pH 11, the observed activity level of the enzyme will most likely be

(A) the same as the level at pH 7

(B) lower than the level at pH 9 ✓

(C) greater than the level at pH 9

(D) between the levels observed at pH 5 and pH 7.

Answer B

Correct. Peroxidase activity drops off as the pH increases beyond 5, indicating that the optimal level was closer to pH 5. The enzyme is likely to be further denatured by a pH greater than 9. Since this enzyme occurs in many organisms, it is reasonable that the optimal pH is closer to neutral than to a strongly basic or strongly acidic pH.

45. In plants the enzyme EPSP synthase catalyzes a reaction between two different substrates, S3P and PEP, to form a product used in the synthesis of three amino acids. The structure of glyphosate, the active chemical in many herbicides, is very similar to that of PEP.

Which of the following explains how glyphosate most likely functions as an herbicide?

(A) Glyphosate prevents synthesis of the three amino acids by competing with PEP for binding to the active site of EPSP synthase. ✓

(B) Glyphosate prevents synthesis of the three amino acids by binding to an allosteric site of EPSP synthase, increasing the likelihood that PEP will bind to the active site.

(C) Glyphosate causes too much synthesis of the three amino acids by reducing the activation energy of the reaction catalyzed by EPSP synthase.

(D) Glyphosate alters the structure of the three amino acids by binding to both S3P and PEP in the active site of EPSP synthase.

Answer A

Correct. Because the structure of glyphosate is very similar to that of PEP, glyphosate acts as a competitive inhibitor of the EPSP enzyme by competing with PEP for the active site of the enzyme.